Integrating Multi-Omics Features for Breast Cancer Classification

Zahin Peerzade & Jessica Vu

SCIENTIFIC PROBLEM

- ★ ER status (ER+ vs ER–) is a crucial clinical marker for breast-cancer treatment and prognosis
- ★ Multi-omics technologies generate thousands of features per patient (RNA, CNV, protein, mutation)
- ★ Challenge: Identify which molecular signals best distinguish ER+ from ER– tumors

OBJECTIVE

Use integrated multi-omics data (RNA expression, copy-number variation, mutation, and protein levels) to predict and interpret breast-cancer molecular subtypes

Dataset

Source: TCGA-BRCA (Kaggle)

Samples: 705 breast-cancer tumors

Features: 1,900 omics variables across four layers (RNA, CNV, mutation, protein)

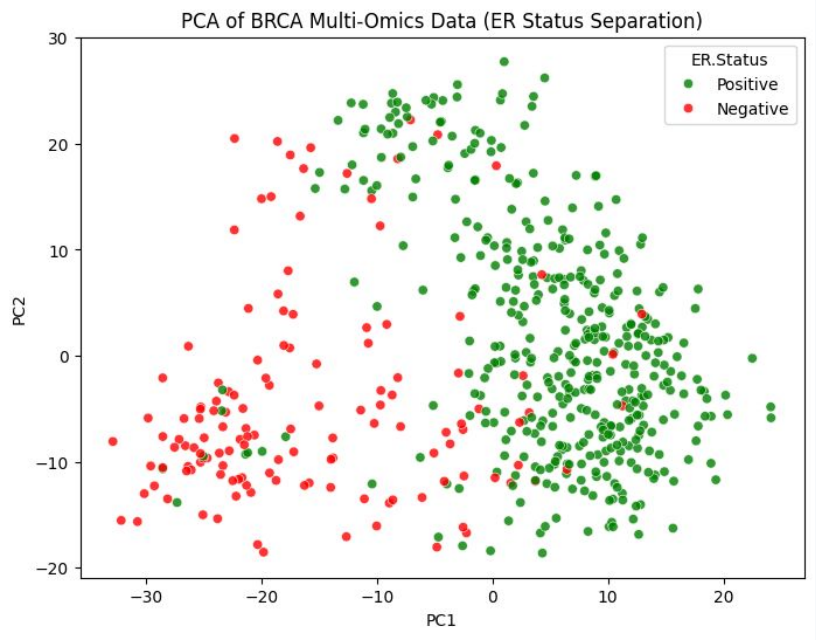
Target variable: ER Status

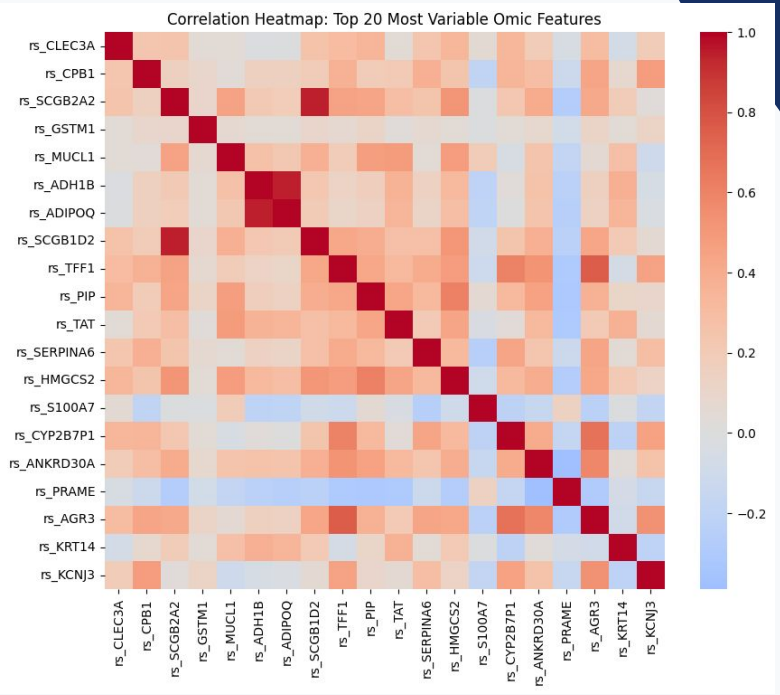
Methodology

- ★ Perform EDA
- ★ Train and compare classification models (Random Forest, Logistic Regression, XGBoost, SVM)
- ★ Evaluate accuracy and feature importance
- ★ Interpret top predictive features across omics layers

Exploratory Data Analysis

549 samples, 1,937 features after preprocessing **(ER + 414, ER – 135)**





PCA shows clustering of ER + vs ER – → biological signal present

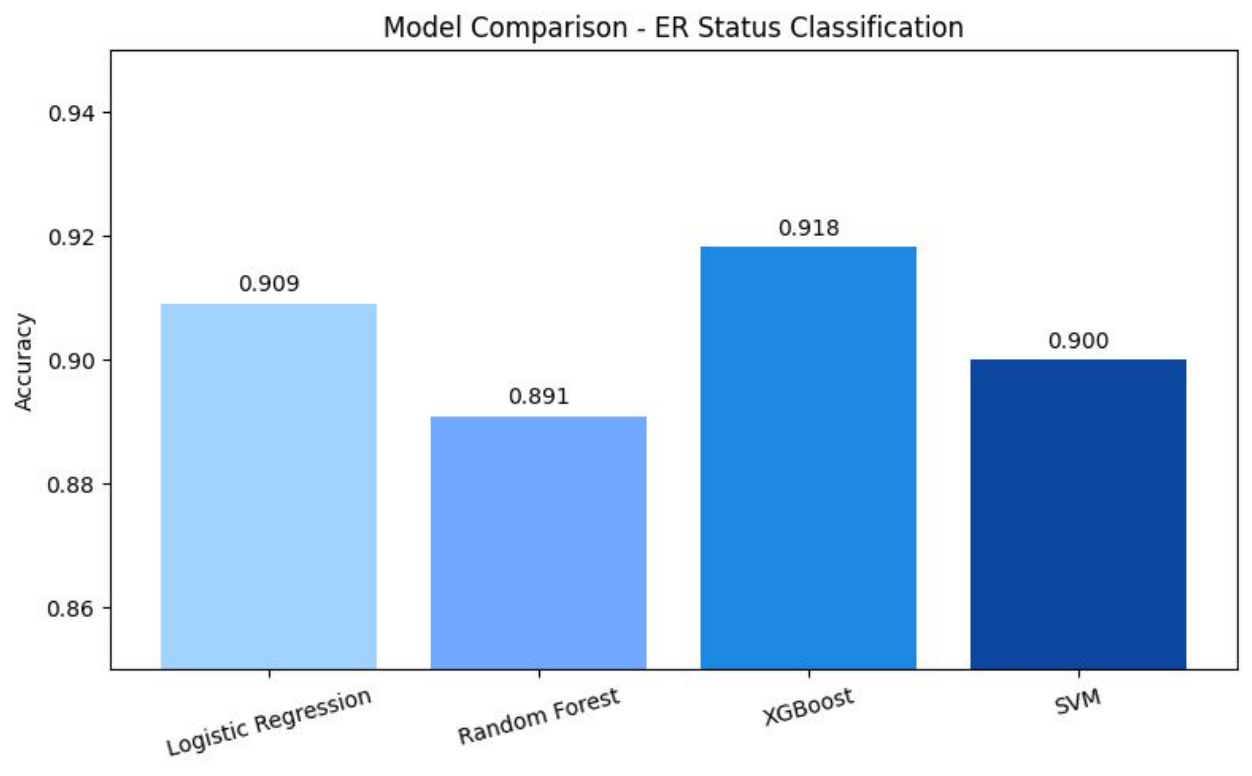
Top 20 feature correlations show mixed positive / negative relations → omics diversity

Workflow

Split: 80 % train / 20 % test
Models: Logistic Regression, Random Forest, XGBoost, SVM
Cross-validated on **5 folds**

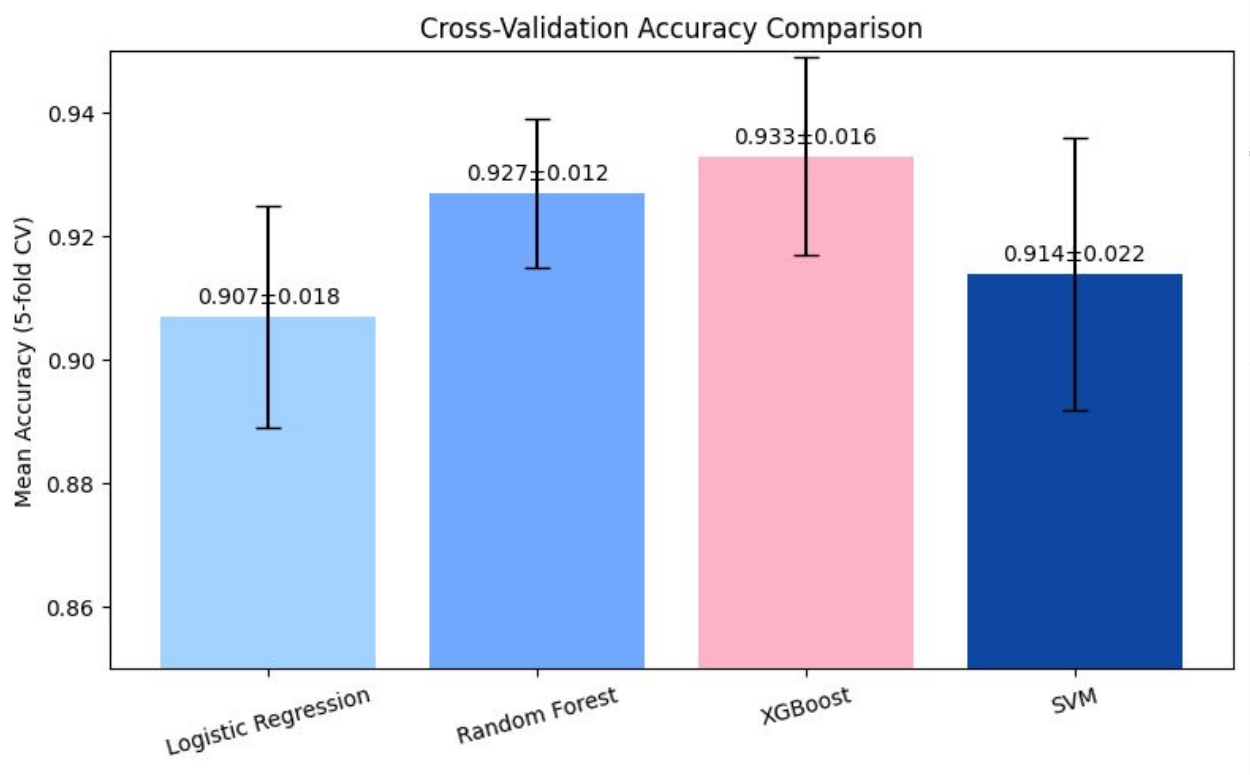
SPLIT —> 8 : 2

Model Performance on Test Set



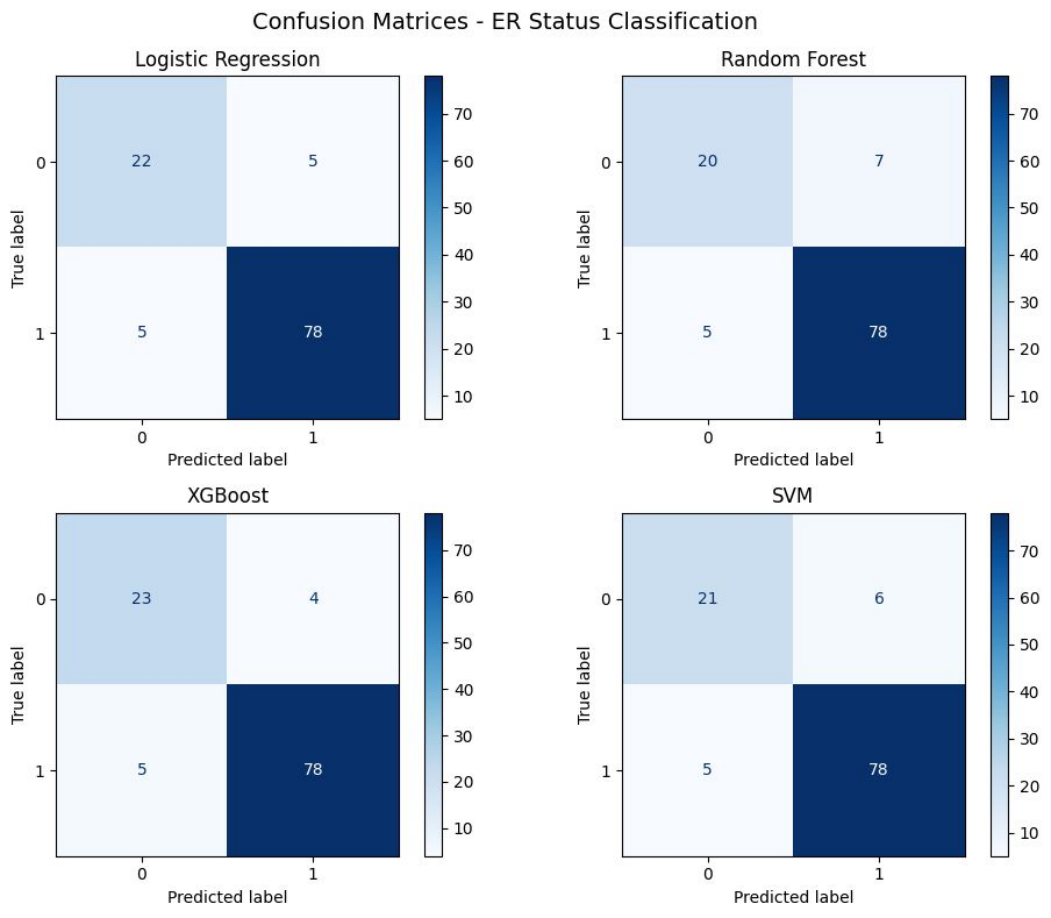
- ★ **XGBoost**: 0.918
- ★ **Logistic Regression**: 0.909
- ★ **SVM**: 0.900
- ★ **Random Forest**: 0.891

Cross Validation Results



- ★ **XGBoost**: 0.933 (+/- 0.016)
- ★ **Logistic Regression**: 0.907 (+/- 0.018)
- ★ **SVM**: 0.914 (+/- 0.022)
- ★ **Random Forest**: 0.927 (+/- 0.012)

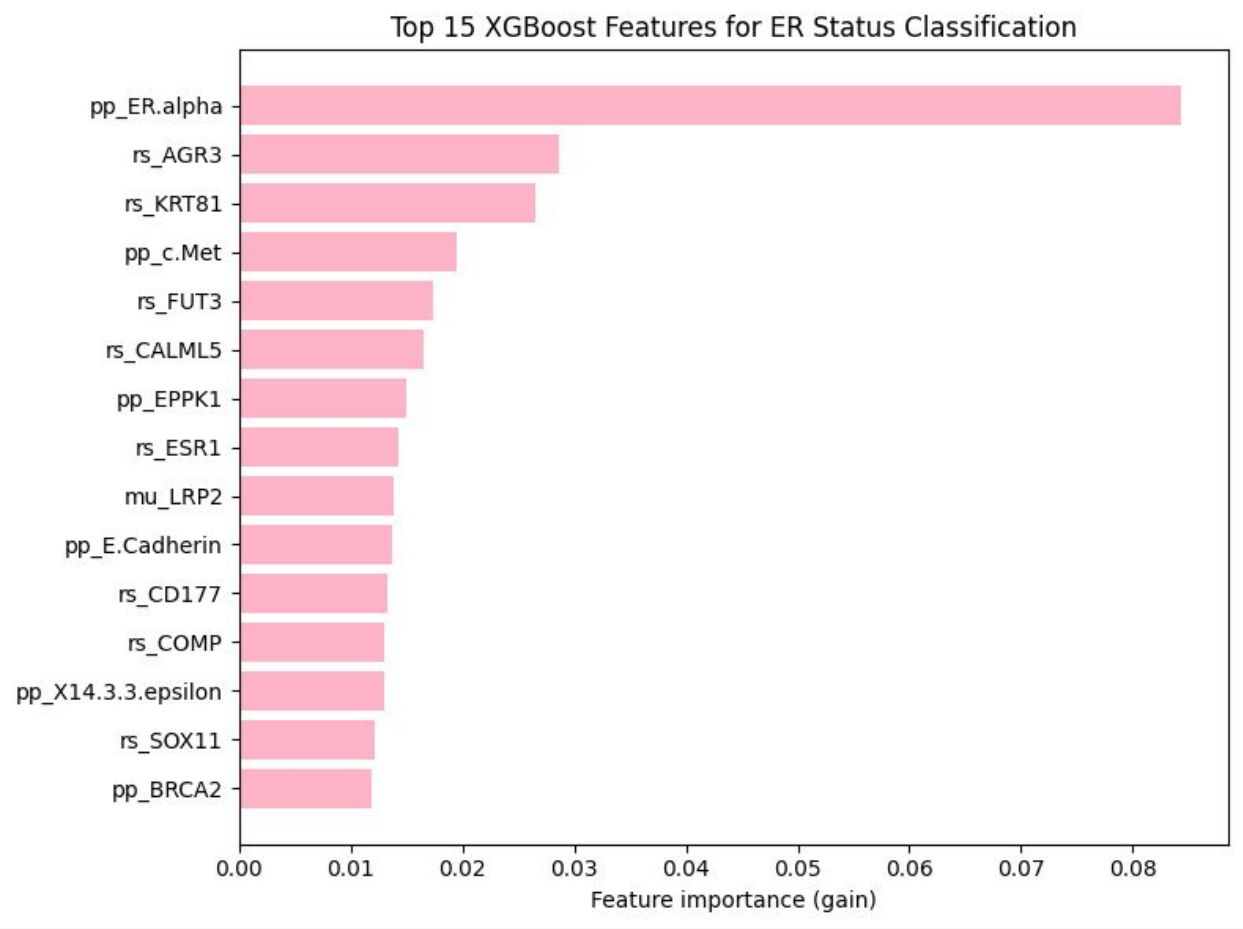
Confusion Matrices



- ★ All models predict ER+ well
- ★ Most errors:
  ER– misclassified as ER+
- ★ XGBoost has lowest false-negatives

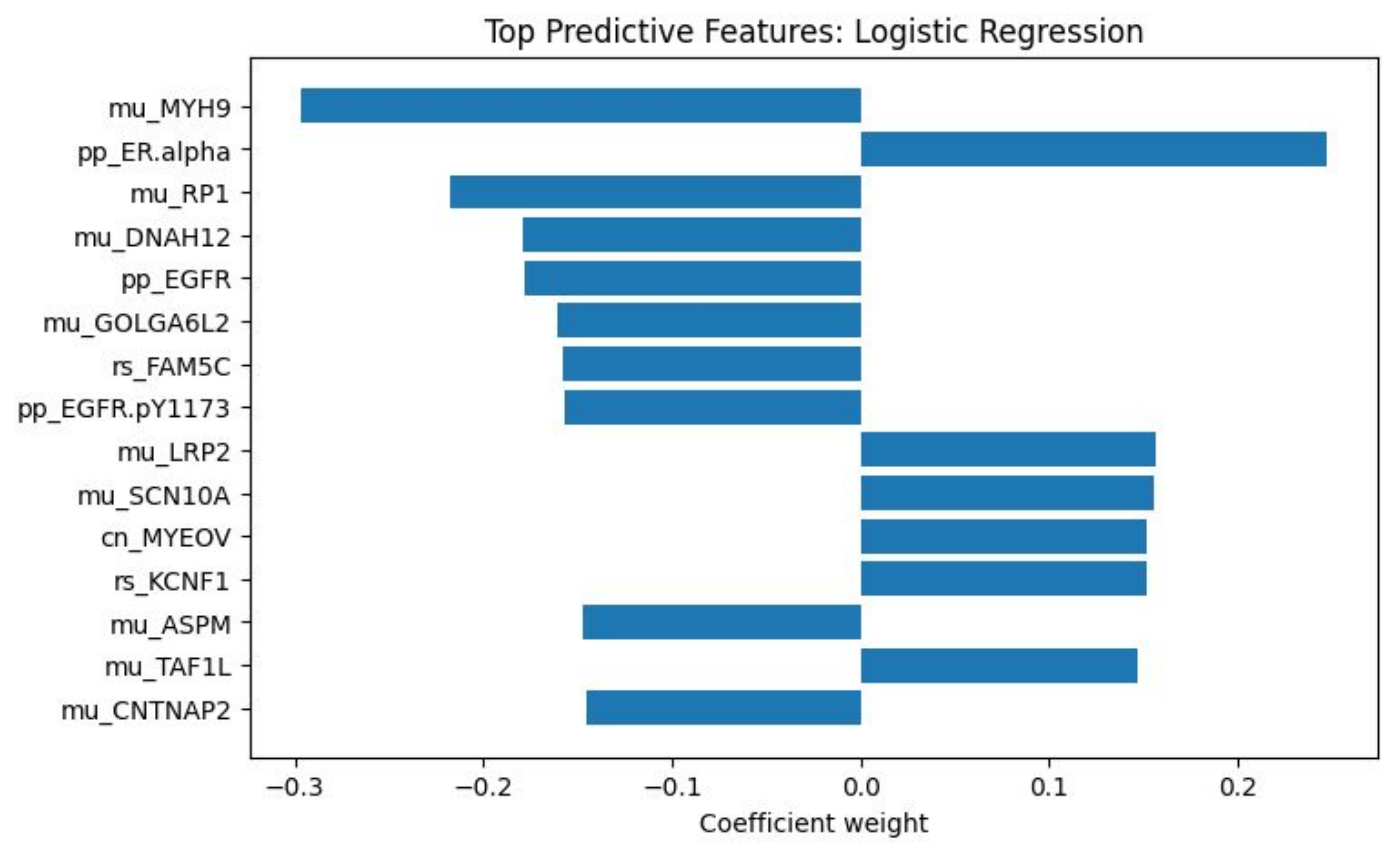
XGBoost Feature Importance

- ★ pp_ER.alpha = strongest signal
- ★ AGR3, KRT81, c-Met also highly predictive
- ★ Highlights protein-level biomarkers



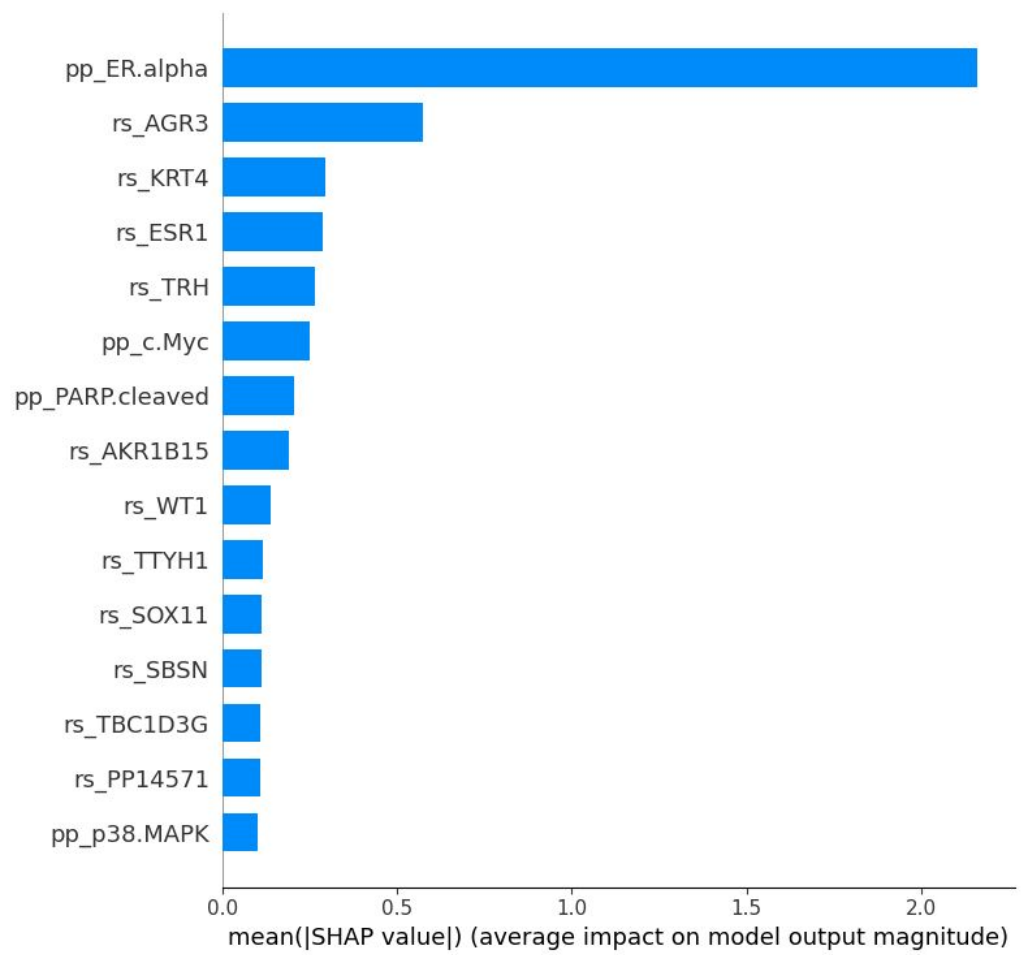
LR Coefficients

- ★ MYH9, ER-alpha show strongest weights
- ★ LR provides feature-level directionality



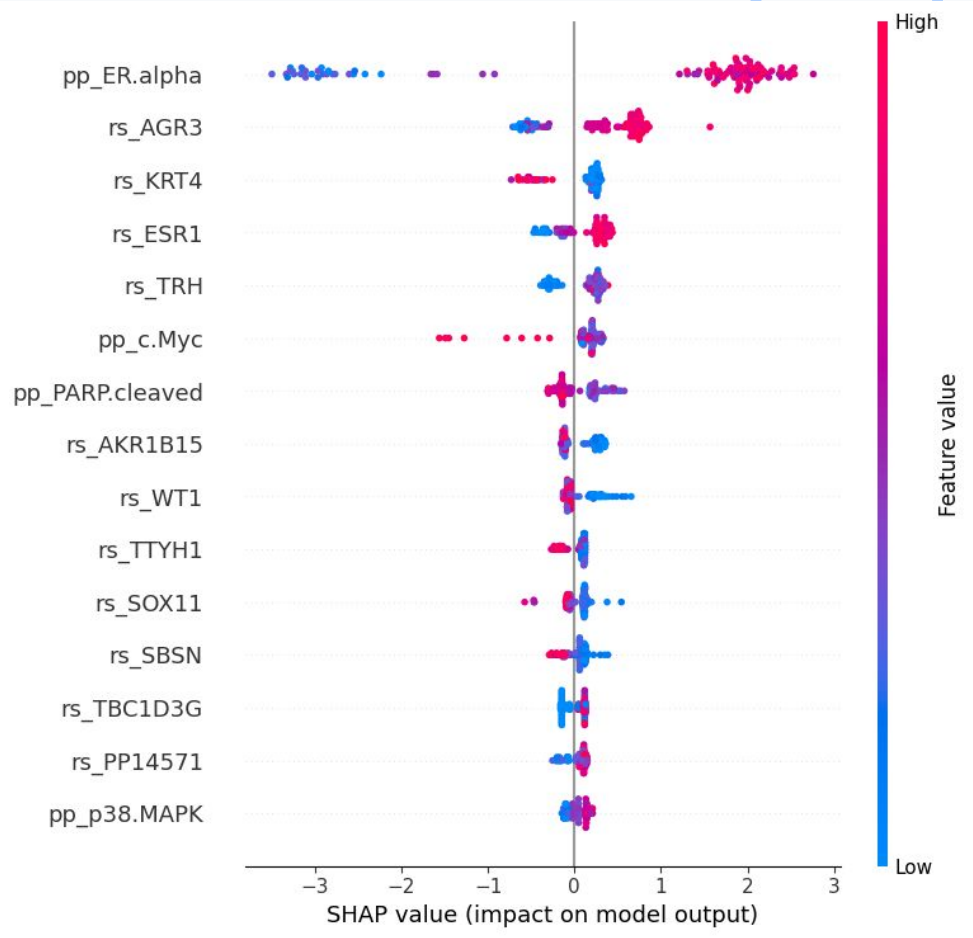
SHAP: Most Influential Features

- ★ pp_ER.alpha dominates model output
- ★ High AGR3, ESR1 push predictions toward ER+
- ★ Feature effects consistent with known ER biology
- ★ SHAP captures non-linear interactions missed by linear models



How each feature drives prediction

- ★ High values (pink) push the model toward ER+
- ★ Low values (blue) push toward ER-
- ★ pp_ER.alpha is the dominant biomarker
- ★ AGR3, ESR1, KRT4 also strongly tilt predictions



SUMMARY

- ★ Multi-omics data provides strong signal for ER classification
- ★ XGBoost is best-performing model on this dataset (test = 0.918, CV = 0.933)
- ★ SHAP highlights biologically meaningful biomarkers
- ★ Omic-layer features (protein + RNA) strongly drive predictions

KEY TAKEAWAYS

- ★ Different ML models emphasize different biological signals (linear vs non-linear).
- ★ SHAP provides deeper interpretability than raw feature importance.
- ★ XGBoost performs best for non-linear multi-omic data.
- ★ ER pathway proteins dominate predictive signal.